

MACHINE LEARNING-BASED LATE DELIVERY RISK PREDICTION IN GLOBAL SUPPLY CHAIN OPERATIONS

Internship Project Report

Submitted in Partial Fulfillment of the Requirements for the Award of

Certificate of the internship

Business Analytics

By

Ashutosh Sharma

MBA (Business Analytics)

Internship Organization

Unified Mentor Pvt. Ltd.

CERTIFICATE

CERTIFICATE OF COMPLETION

This is to certify that the internship project entitled

"Machine Learning-Based Late Delivery Risk Prediction in Global Supply Chain Operations"

has been successfully completed by

Ashutosh Sharma

as a part of the Business Analytics Internship conducted by **Unified Mentor Pvt. Ltd.**

The project demonstrates the application of Business Analytics, Data Analytics, Machine Learning, Predictive Analytics, and Dashboard Development techniques for solving a real-world logistics problem.

The work presented in this report is original and completed during the internship period under proper guidance.

Signature

Project Mentor

Unified Mentor Pvt. Ltd.

DECLARATION

DECLARATION

I hereby declare that this internship project report entitled

"Machine Learning-Based Late Delivery Risk Prediction in Global Supply Chain Operations"

submitted in partial fulfillment of the requirements for the degree of **Master of Business Administration (Business Analytics)** is my original work.

The work presented in this report has been carried out during my Business Analytics Internship at **Unified Mentor Pvt. Ltd.**

The information and results presented in this report have not been submitted previously for the award of any degree, diploma, or certificate.

All sources of information have been properly acknowledged.

Ashu Pandit

MBA (Business Analytics)

Amity University Online

Date:

Place:

ACKNOWLEDGEMENT

I express my sincere gratitude to **Unified Mentor Pvt. Ltd.** for providing me with the opportunity to undertake this Business Analytics Internship and work on a real-world Machine Learning project.

This internship has significantly enhanced my understanding of Business Analytics, Data Science, Machine Learning, Predictive Analytics, and Dashboard Development. The guidance, support, and valuable learning opportunities provided throughout the internship have contributed immensely to my professional growth.

I extend my heartfelt appreciation to all the mentors, faculty members, and the open-source community behind Python, Scikit-Learn, Streamlit, Pandas, NumPy, Plotly, and Matplotlib for developing the tools and libraries used in this project.

Finally, I thank my family and friends for their continuous encouragement, motivation, and support throughout the completion of this internship project.

EXECUTIVE SUMMARY

The logistics industry faces significant challenges due to shipment delays, which directly impact customer satisfaction, operational efficiency, and overall business performance. Traditional logistics systems often identify delivery issues only after delays have already occurred, leaving limited opportunities for proactive intervention.

This internship project presents an end-to-end Machine Learning-based solution designed to predict late delivery risk before shipment dispatch. The project was completed as part of the Business Analytics Internship at **Unified Mentor Pvt. Ltd.**

The project follows the complete Business Analytics lifecycle beginning with data collection, preprocessing, exploratory data analysis (EDA), feature engineering, predictive modeling, model evaluation, and dashboard development.

The dataset contains **180,519 shipment records** with **32 operational features**, including customer information, shipping details, order information, market data, product information, and logistics-related variables. Multiple Machine Learning algorithms including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost were trained and evaluated.

The selected model achieved:

- Accuracy: **73.27%**
- Precision: **84.28%**
- Recall: **62.99%**
- F1 Score: **72.10%**
- ROC-AUC Score: **0.8156**

A professional multi-page Streamlit dashboard was developed to provide logistics managers with real-time business insights. The dashboard includes an Executive Dashboard, Risk Prediction module, Risk Analytics, Model Performance analysis, Operations Center, and About Project section.

The proposed solution enables logistics managers to identify high-risk shipments before dispatch, improve operational planning, allocate resources effectively, and enhance customer satisfaction through proactive decision-making.

Overall, this project demonstrates the practical application of Business Analytics, Machine Learning, and Data Visualization techniques in solving a real-world logistics problem and highlights the value of predictive analytics in modern supply chain management.

TABLE OF CONTENTS

1. Introduction
 2. Company Profile
 3. Problem Statement
 4. Project Objectives
 5. Literature Review
 6. Dataset Description
 7. Data Preprocessing
 8. Exploratory Data Analysis
 9. Feature Engineering
 10. Machine Learning Model Development
 11. Model Evaluation
 12. Streamlit Dashboard Development
 13. Business Recommendations
 14. Conclusion
 15. Future Scope
 16. References
 17. Appendix
-

LIST OF FIGURES

Figure 1 Project Workflow

Figure 2 Data Preprocessing Pipeline

Figure 3 Machine Learning Workflow

Figure 4 Dashboard Architecture

Figure 5 Late Delivery Distribution

Figure 6 Shipping Mode Analysis

Figure 7 Market Distribution

Figure 8 Customer Segment Analysis

Figure 9 ROC Curve

Figure 10 Confusion Matrix

LIST OF TABLES

Table 1 Dataset Description

Table 2 Feature Engineering

Table 3 Model Comparison

Table 4 Model Performance

Table 5 Technology Stack

Table 6 Dashboard Features

CHAPTER 2

INTRODUCTION

2.1 Introduction

The rapid growth of globalization, e-commerce, and international trade has significantly increased the complexity of supply chain operations. Organizations operating across multiple countries must ensure that products are delivered efficiently while maintaining customer satisfaction and minimizing operational costs. Logistics companies play a critical role in managing transportation, warehousing, inventory, and order fulfillment to ensure that goods reach customers on time.

One of the major challenges faced by logistics organizations is **late delivery**. Shipment delays can occur due to numerous factors such as transportation issues, weather conditions, route congestion, warehouse delays, incorrect planning, supply shortages, customs clearance, and operational inefficiencies. These delays increase operational costs, affect customer satisfaction, and damage an organization's reputation.

Traditionally, logistics companies identify delivery delays only after they have already occurred. This reactive approach limits the organization's ability to take corrective actions before the shipment reaches the customer. As businesses become increasingly data-driven, organizations

are adopting predictive analytics and machine learning techniques to forecast potential delivery delays before shipment dispatch.

Machine Learning enables organizations to analyze historical shipment data, identify hidden patterns, and estimate the probability of delivery delays. By predicting high-risk shipments in advance, logistics managers can allocate additional resources, adjust transportation strategies, communicate proactively with customers, and improve overall operational performance.

This internship project focuses on developing an end-to-end Machine Learning-Based Late Delivery Risk Prediction System for APL Logistics. The project integrates data preprocessing, exploratory data analysis, feature engineering, predictive modeling, and dashboard development into a unified Business Analytics solution.

The project demonstrates how modern analytics techniques can transform raw logistics data into actionable business insights that support proactive decision-making and improve supply chain efficiency.

2.2 Company Profile

Unified Mentor Pvt. Ltd.

Unified Mentor Pvt. Ltd. is an educational and professional training organization that provides industry-oriented internship programs and practical learning opportunities in emerging technology domains. The organization aims to bridge the gap between academic learning and industry requirements by providing hands-on experience through real-world projects.

The Business Analytics Internship offered by Unified Mentor focuses on developing practical skills in data analytics, business intelligence, machine learning, dashboard development, and predictive analytics. Interns work on industry-inspired case studies that require the application of statistical analysis, programming, data visualization, and machine learning techniques to solve business problems.

During the internship, participants gain exposure to the complete analytics lifecycle, including:

- Data Collection
- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Machine Learning Model Development
- Model Evaluation

- Dashboard Development
- Business Reporting
- Deployment of Analytical Solutions

The internship emphasizes project-based learning, enabling participants to develop technical expertise while strengthening analytical thinking and business problem-solving skills.

This project represents the practical application of the concepts learned during the internship and demonstrates the integration of Business Analytics methodologies with modern Machine Learning technologies.

2.3 Industry Overview

The logistics and supply chain industry forms the backbone of global commerce. Efficient transportation and timely delivery of goods are essential for manufacturing, retail, healthcare, e-commerce, and international trade. Organizations invest significant resources in optimizing supply chain operations to reduce costs, improve delivery performance, and enhance customer satisfaction.

Recent advancements in Business Analytics, Artificial Intelligence, and Machine Learning have transformed logistics management. Companies increasingly rely on predictive analytics to forecast shipment delays, optimize routes, manage inventory, and improve resource allocation.

Key challenges faced by logistics organizations include:

- Delivery delays
- Rising transportation costs
- Inventory management
- Demand uncertainty
- Customer expectations
- Global supply chain disruptions
- Route optimization
- Operational inefficiencies

Predictive analytics provides organizations with the ability to identify potential risks before they occur, enabling proactive operational planning and improved decision-making.

2.4 Problem Statement

Late deliveries are among the most critical operational challenges in global logistics. Delayed shipments result in financial losses, reduced customer satisfaction, inefficient resource utilization, and disruptions throughout the supply chain.

Traditional logistics systems primarily rely on historical reporting and manual monitoring, making it difficult to identify high-risk shipments before dispatch. As shipment volumes continue to increase, manual risk assessment becomes inefficient and prone to error.

The primary problem addressed in this project is:

How can Machine Learning be used to accurately predict shipment delays before dispatch and support proactive logistics decision-making?

To address this problem, this project develops a predictive analytics system capable of identifying high-risk shipments using historical operational data and presenting the results through an interactive business dashboard.

2.5 Project Objectives

The project has the following objectives:

Primary Objectives

- Predict late delivery risk using Machine Learning.
- Identify high-risk shipments before dispatch.
- Develop an interactive dashboard for logistics managers.
- Improve decision-making through predictive analytics.

Secondary Objectives

- Perform exploratory data analysis.
 - Engineer meaningful business features.
 - Compare multiple Machine Learning models.
 - Evaluate model performance using industry-standard metrics.
 - Deploy the solution using Streamlit Cloud.
 - Present insights through interactive visualizations.
-

2.6 Scope of the Project

The scope of this project includes the complete Business Analytics workflow required for developing a predictive logistics solution.

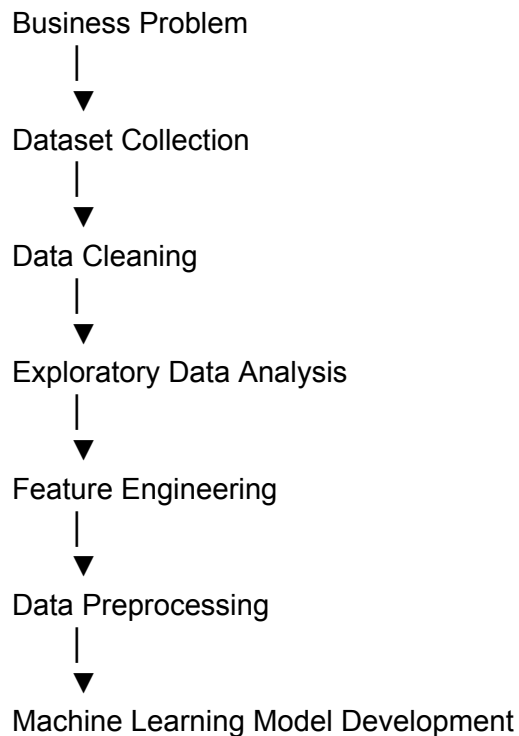
The project covers:

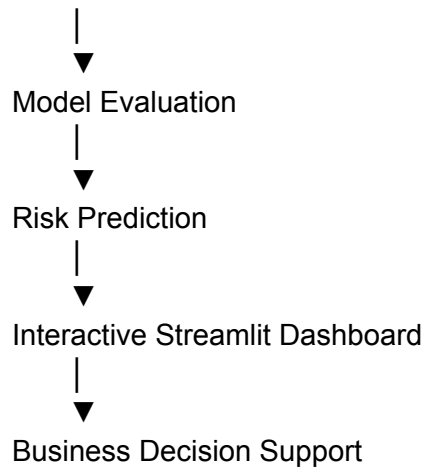
- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering
- Machine Learning model development
- Model evaluation
- Dashboard development
- Deployment on Streamlit Cloud
- Business recommendations

However, the project does not include real-time shipment tracking, live integration with logistics management systems, or continuous model retraining using streaming data. These aspects are identified as potential future enhancements.

2.7 Project Workflow

The project follows a structured Business Analytics lifecycle:





CHAPTER 3

LITERATURE REVIEW

3.1 Introduction

The logistics and supply chain industry has experienced significant transformation due to advancements in Business Analytics, Artificial Intelligence (AI), and Machine Learning (ML). Organizations are increasingly adopting predictive analytics to optimize logistics operations, reduce transportation costs, improve customer satisfaction, and enhance operational efficiency.

Machine Learning enables organizations to analyze large volumes of historical operational data and identify hidden patterns that support proactive decision-making. In logistics, predictive models are widely used for shipment delay prediction, route optimization, demand forecasting, inventory management, warehouse optimization, and transportation planning.

This chapter reviews existing research related to predictive analytics, supply chain management, machine learning algorithms, dashboard development, and business intelligence systems.

3.2 Business Analytics in Logistics

Business Analytics is the process of transforming raw data into meaningful information that supports organizational decision-making. Modern logistics companies generate massive volumes of operational data from transportation systems, warehouse management systems, customer transactions, and shipment tracking applications.

Business Analytics enables organizations to:

- Improve operational efficiency
- Reduce transportation costs
- Optimize inventory
- Improve delivery performance
- Support strategic planning
- Increase customer satisfaction

In logistics organizations, Business Analytics supports managers by providing actionable insights derived from operational data rather than relying solely on intuition or historical reports.

The increasing availability of large logistics datasets has made predictive analytics one of the most valuable tools for supply chain optimization.

3.3 Machine Learning in Supply Chain Management

Machine Learning has become one of the most influential technologies in modern supply chain management.

Unlike traditional rule-based systems, Machine Learning algorithms learn from historical data and continuously improve prediction accuracy.

Machine Learning applications in logistics include:

- Shipment delay prediction
- Demand forecasting
- Inventory optimization
- Warehouse automation
- Route optimization
- Vehicle scheduling
- Customer segmentation
- Fraud detection

Predictive Machine Learning models enable organizations to identify operational risks before they occur, allowing managers to take preventive actions rather than reactive measures.

In this project, Machine Learning is applied specifically to predict the probability of shipment delays before dispatch.

3.4 Predictive Analytics

Predictive Analytics combines statistics, machine learning, and historical data analysis to estimate future events.

The objective of predictive analytics is not simply to understand historical trends but to forecast future outcomes.

Predictive analytics follows several stages:

1. Data Collection
2. Data Cleaning
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. Prediction
7. Business Decision Support

The logistics industry increasingly relies on predictive analytics to improve operational planning and reduce uncertainty.

By predicting shipment delays before dispatch, organizations can allocate resources more effectively and improve customer communication.

3.5 Classification Algorithms

Classification is one of the most common supervised Machine Learning techniques.

The objective of classification is to predict whether an observation belongs to a particular category.

In this project, the target variable is:

Late Delivery Risk

Possible classes include:

- Late Delivery
- On-Time Delivery

Several classification algorithms were evaluated.

Logistic Regression

Logistic Regression is a statistical classification algorithm widely used for binary classification problems.

Advantages include:

- Easy interpretation
- Fast training
- Low computational cost
- Strong baseline performance

However, Logistic Regression may struggle when relationships between variables become highly nonlinear.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Advantages:

- High accuracy
- Reduced overfitting
- Handles nonlinear relationships
- Works well with large datasets

Random Forest is widely used in logistics analytics due to its robustness and predictive performance.

Gradient Boosting

Gradient Boosting builds multiple decision trees sequentially, where each new tree attempts to reduce the errors of previous trees.

Advantages:

- Excellent predictive performance
- Handles complex datasets
- Good generalization ability

The algorithm is particularly effective for structured business datasets.

XGBoost

Extreme Gradient Boosting (XGBoost) is an optimized implementation of gradient boosting.

Advantages:

- Faster training
- Better regularization
- High predictive accuracy
- Handles missing values efficiently

XGBoost is widely recognized as one of the most powerful algorithms for structured tabular datasets.

3.6 Feature Engineering

Feature Engineering is the process of creating new variables that improve model performance.

Instead of relying only on original dataset variables, engineered features capture additional business knowledge.

Examples created in this project include:

- Shipping Pressure Index
- Mode Risk Flag
- Order Complexity Score
- Regional Congestion

These engineered features improved the model's ability to distinguish between high-risk and low-risk shipments.

3.7 Model Evaluation Metrics

Evaluating Machine Learning models requires multiple performance metrics.

The following metrics were used:

Accuracy

Measures the percentage of correctly classified observations.

Precision

Measures how many predicted delayed shipments were actually delayed.

Recall

Measures how effectively the model identifies delayed shipments.

F1 Score

Provides a balanced measure of Precision and Recall.

ROC-AUC

Measures the model's ability to distinguish between delayed and non-delayed shipments across different classification thresholds.

These metrics provide a comprehensive understanding of model performance.

3.8 Business Intelligence Dashboards

Business Intelligence dashboards transform analytical results into visual insights.

Interactive dashboards allow managers to:

- Monitor Key Performance Indicators (KPIs)
- Analyze operational trends
- Compare model performance
- Explore shipment risks
- Support decision-making

In this project, Streamlit was used to develop an interactive dashboard consisting of:

- Executive Dashboard
- Risk Prediction
- Risk Analytics
- Model Performance
- Operations Center
- About Project

The dashboard serves as a decision-support tool for logistics managers.

3.9 Research Gap

Although numerous studies have explored logistics analytics and shipment delay prediction, many focus primarily on predictive modeling without integrating business intelligence dashboards for operational decision support.

Furthermore, several studies evaluate only a single machine learning algorithm, limiting comparative analysis.

This project addresses these gaps by:

- Comparing multiple machine learning models.
 - Performing comprehensive feature engineering.
 - Developing an interactive Streamlit dashboard.
 - Providing business recommendations based on predicted shipment risk.
 - Integrating predictive analytics with business intelligence.
-

3.10 Chapter Summary

This chapter reviewed the theoretical concepts and previous research related to Business Analytics, Machine Learning, Predictive Analytics, Classification Algorithms, Feature Engineering, and Business Intelligence.

The literature indicates that predictive analytics plays a significant role in improving logistics operations through proactive risk identification and operational planning.

The insights gained from the literature guided the design and implementation of the Machine Learning-Based Late Delivery Risk Prediction System developed in this internship project.

CHAPTER 4

DATASET DESCRIPTION AND RESEARCH METHODOLOGY

4.1 Introduction

The success of any Machine Learning model depends largely on the quality of the data used for training and evaluation. Since logistics operations generate vast amounts of operational data, proper preprocessing and feature engineering are essential for extracting meaningful patterns and improving predictive performance.

This chapter describes the dataset used in the project, the preprocessing techniques applied, the exploratory data analysis performed, and the complete methodology followed to develop the Machine Learning-Based Late Delivery Risk Prediction System.

4.2 Dataset Description

The dataset used in this project represents historical shipment records from logistics operations. It contains information related to customers, products, shipping methods, order details, regional markets, and financial performance.

The primary objective is to predict whether an order is likely to experience a delivery delay before shipment.

Dataset Summary

Attribute	Value
Total Records	180,519
Total Features	32
Target Variable	Late_delivery_risk
Problem Type	Binary Classification
Domain	Supply Chain & Logistics
Project Type	Predictive Analytics

The dataset contains both numerical and categorical variables, making it suitable for supervised Machine Learning classification.

4.3 Dataset Features

The dataset includes information related to customer demographics, product characteristics, sales information, shipping details, and operational metrics.

Some important variables include:

- Type
- Days for Shipment (Scheduled)
- Benefit per Order
- Sales per Customer
- Category Name
- Customer City
- Customer Country
- Customer Segment
- Customer State
- Department Name
- Market
- Order City
- Order Country
- Order Item Discount
- Order Item Discount Rate
- Order Item Product Price

- Order Item Profit Ratio
- Order Item Quantity
- Sales
- Order Item Total
- Order Profit Per Order
- Order Region
- Order State
- Order Status
- Product Name
- Product Price
- Shipping Mode
- Shipping Pressure Index (*Engineered*)
- Mode Risk Flag (*Engineered*)
- Order Complexity Score (*Engineered*)
- Regional Congestion (*Engineered*)

The target variable **Late_delivery_risk** indicates whether a shipment is expected to be delayed.

4.4 Research Methodology

The project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** framework, which is widely used in Business Analytics and Machine Learning projects.

The methodology consists of the following phases:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Exploratory Data Analysis
5. Feature Engineering
6. Model Development
7. Model Evaluation
8. Deployment
9. Business Recommendations

This structured methodology ensures a systematic and reproducible approach to solving the business problem.

4.5 Data Collection

The dataset consists of historical shipment records collected from logistics operations. It contains transactional and operational information used to analyze shipment performance and delivery outcomes.

The data includes:

- Customer Information
- Product Information
- Sales Information
- Shipping Details
- Regional Information
- Order Information
- Operational Metrics

The collected dataset serves as the foundation for predictive modeling and business analytics.

4.6 Data Preprocessing

Raw datasets often contain inconsistencies, missing values, duplicate records, and variables that require transformation before analysis.

The following preprocessing techniques were applied:

Data Cleaning

- Removed duplicate records
- Verified data integrity
- Checked data types
- Standardized categorical values

Missing Value Handling

The dataset was examined for missing values. Appropriate preprocessing ensured that the final dataset was suitable for model training.

Feature Selection

Irrelevant variables were excluded from model training while important predictive variables were retained.

Encoding

Categorical variables were transformed into numerical representations using Scikit-Learn preprocessing techniques to make them suitable for machine learning algorithms.

Data Transformation

Numerical variables were standardized where required using preprocessing pipelines.

These preprocessing steps improved data quality and enhanced the predictive performance of the models.

4.7 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify important relationships between variables.

The analysis included:

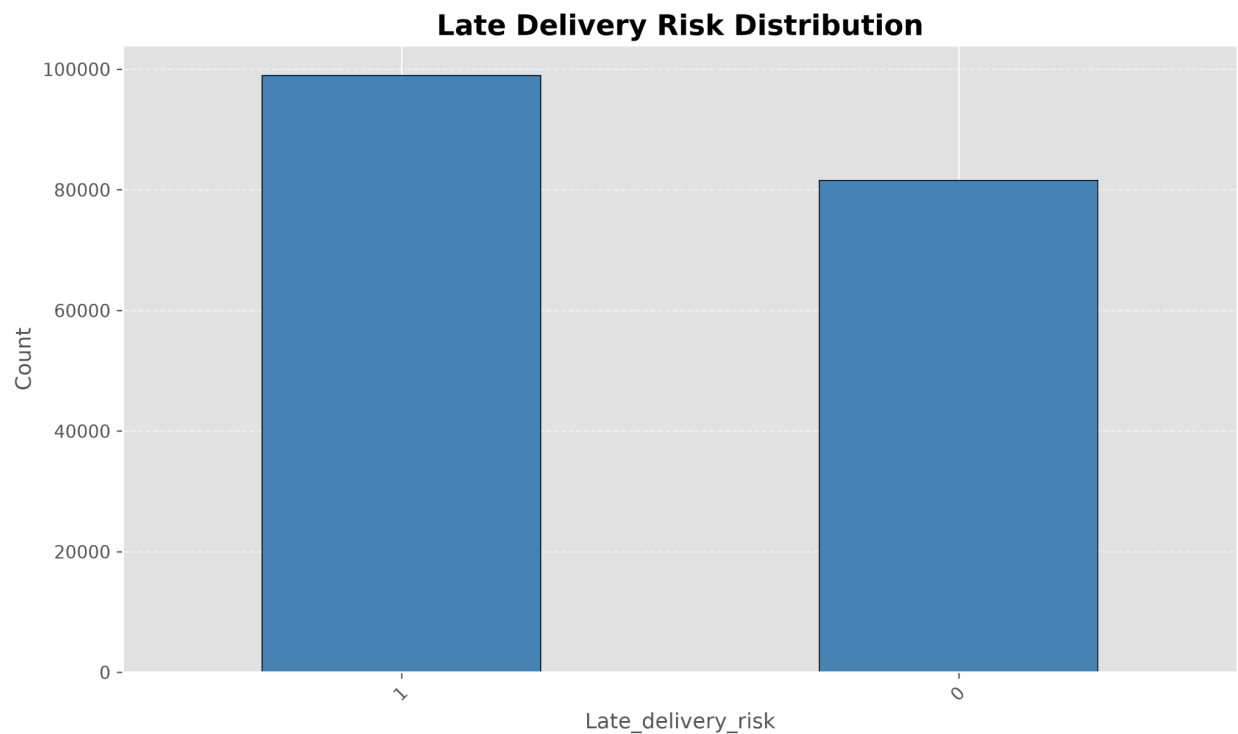
- Distribution of shipment delays
- Shipping mode analysis
- Customer segment analysis
- Market distribution
- Regional analysis
- Sales analysis
- Product analysis
- Correlation analysis

EDA helped identify important trends and patterns that guided feature engineering and model development.

4.8 Visual Analysis

Several charts and visualizations were generated during the analysis phase.

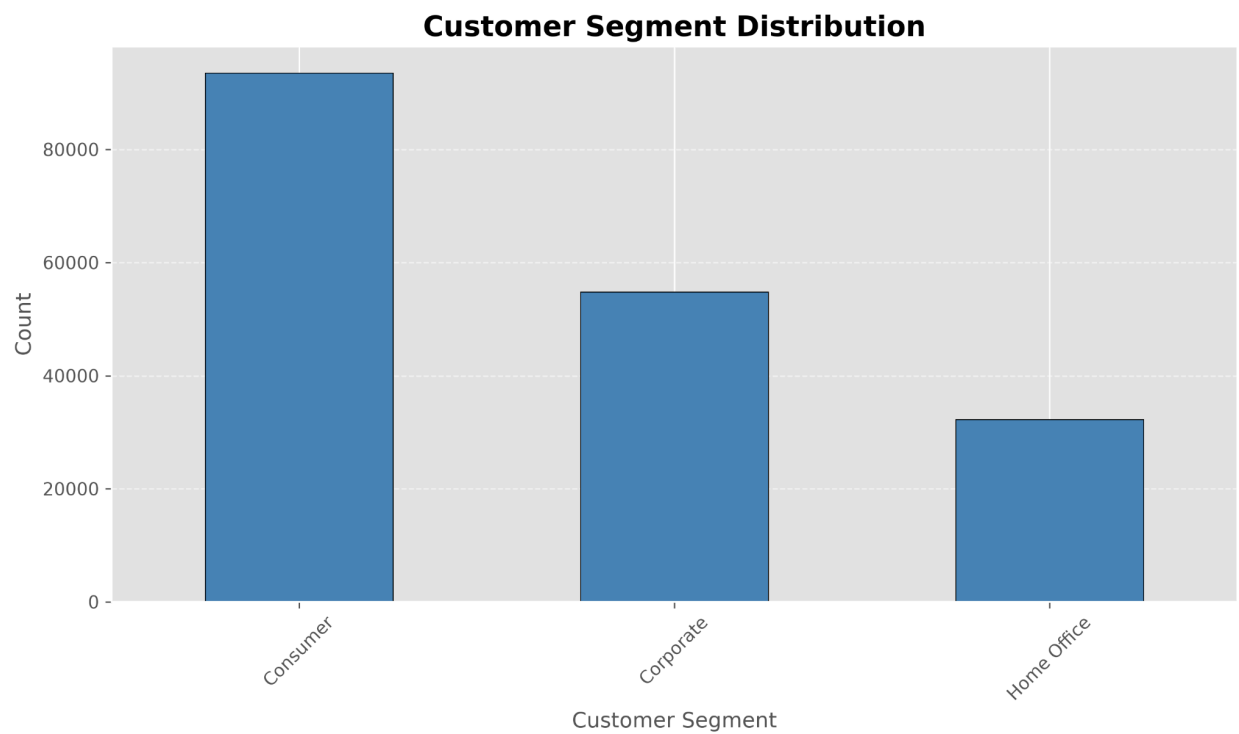
Include your generated charts here, for example:



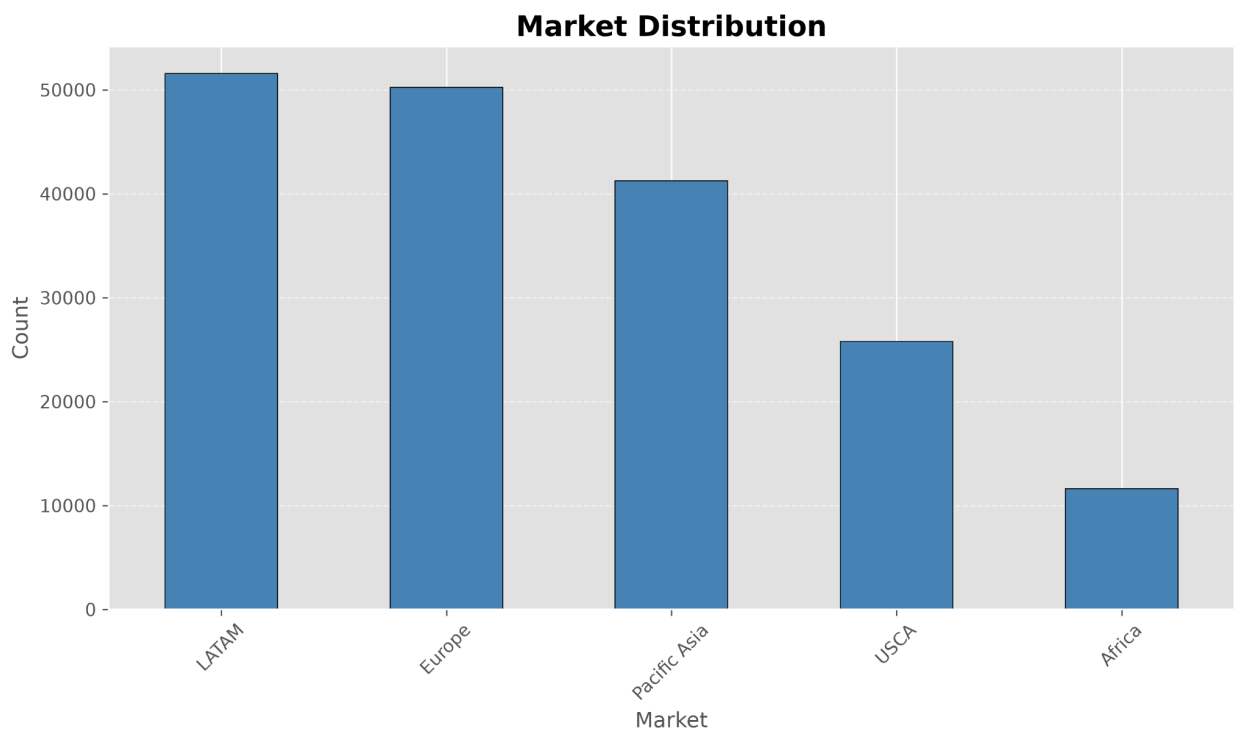
● Figure 4.1 — Late Delivery Distribution



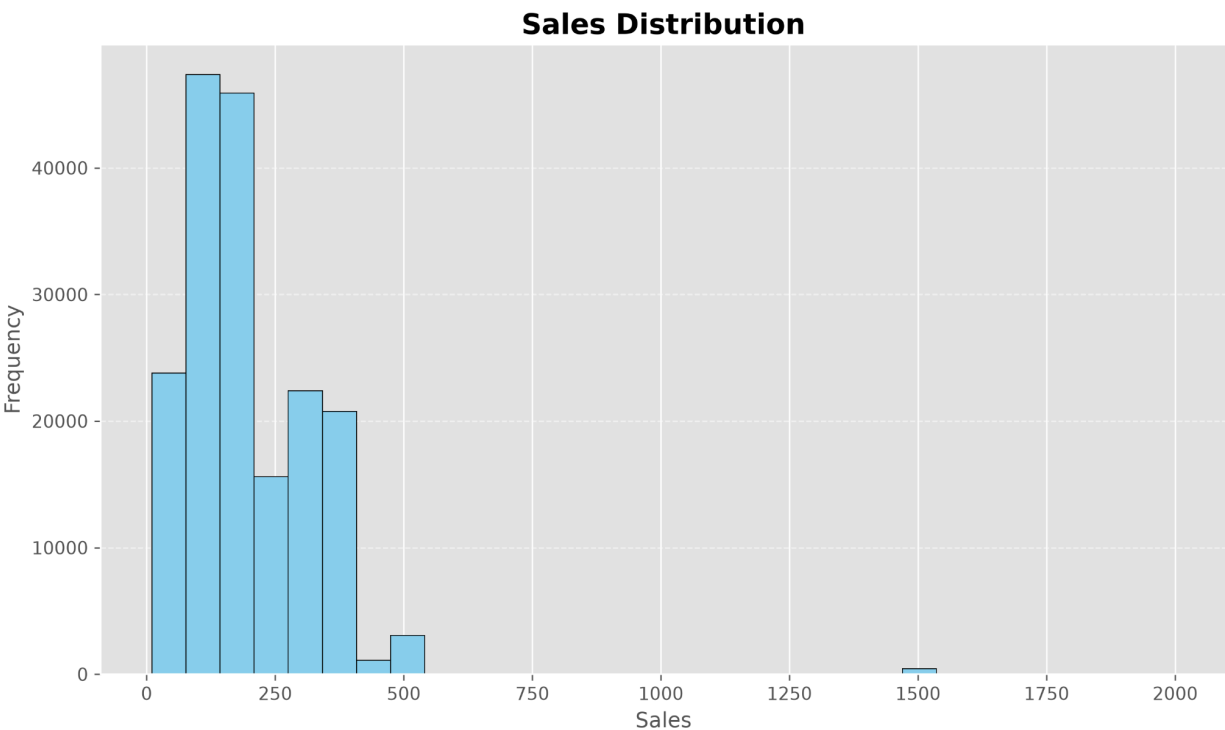
● Figure 4.2 — Shipping Mode Distribution



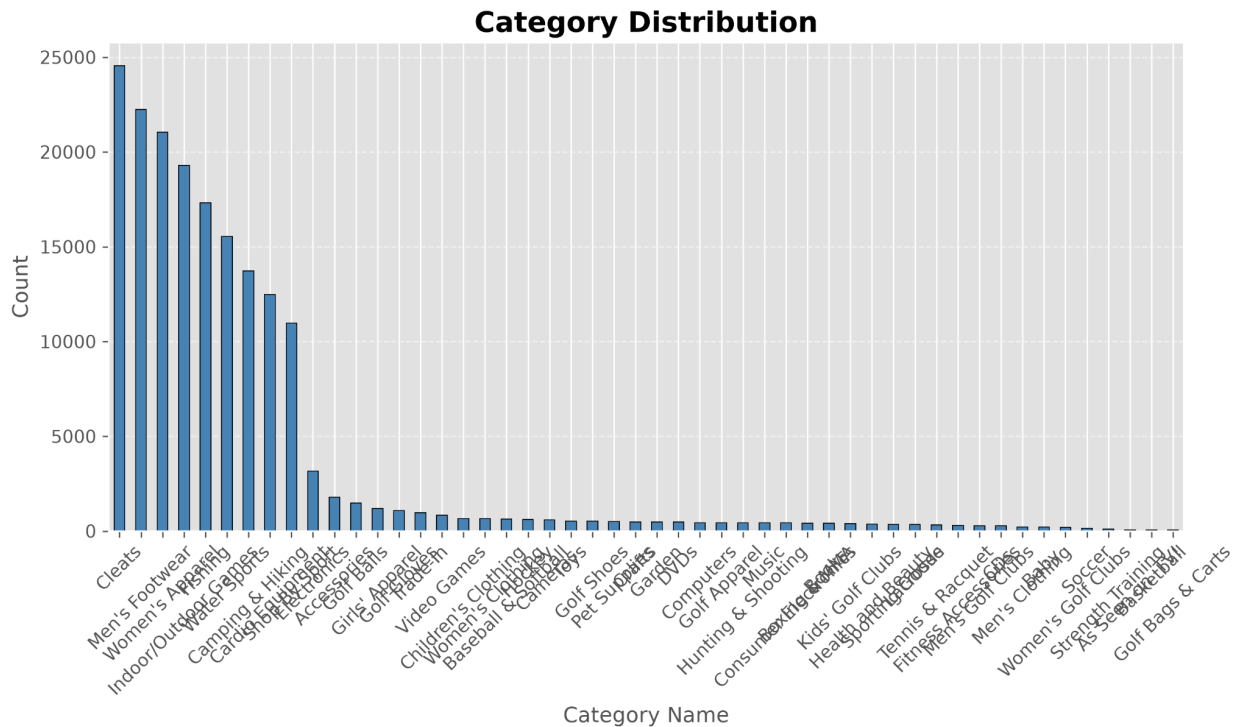
● Figure 4.3 — Customer Segment Analysis



- Figure 4.4 — Market Distribution



- Figure 4.5 — Sales Distribution



- Figure 4.6 — Product Category Analysis

4.9 Feature Engineering

To improve model performance, several business-specific features were engineered.

Shipping Pressure Index

This feature measures shipment pressure by considering the ratio between order quantity and scheduled shipment days.

It helps identify orders with high shipping workload relative to the available delivery time.

Mode Risk Flag

This binary feature identifies whether the shipment uses **Standard Class**, which may have a higher likelihood of delays compared to faster shipping methods.

Order Complexity Score

This feature combines:

- Order Quantity
- Product Price
- Discount Rate

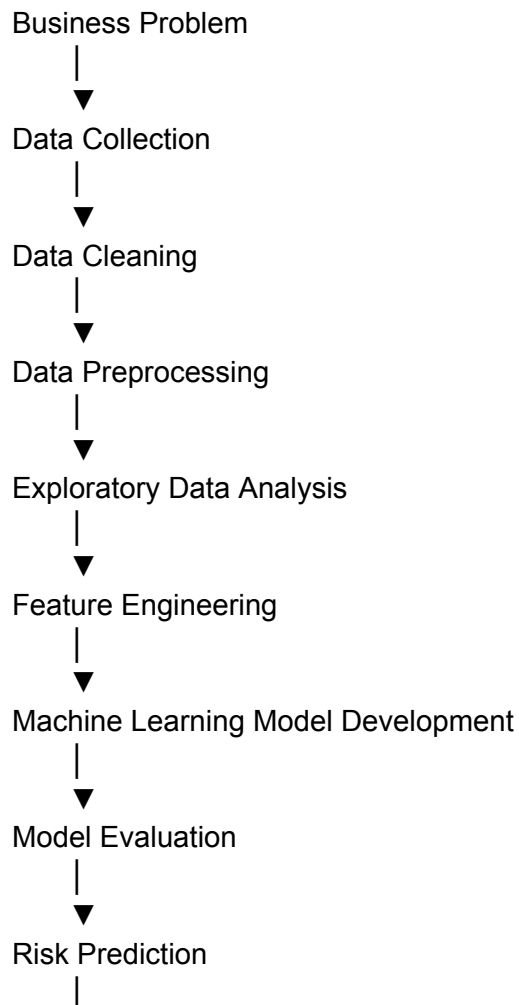
to estimate the complexity of fulfilling an order.

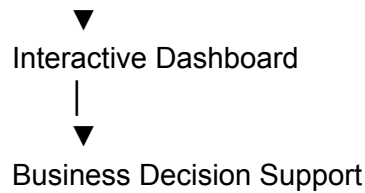
Regional Congestion

This engineered feature represents regional logistics congestion and helps capture operational variations across different markets.

Feature engineering significantly improved the predictive capability of the machine learning models by incorporating domain-specific knowledge.

4.10 Methodology Workflow





4.11 Chapter Summary

This chapter presented the dataset description and the research methodology adopted for the project. It described the characteristics of the logistics dataset, preprocessing techniques, exploratory data analysis, feature engineering, and the overall workflow followed for developing the Machine Learning-Based Late Delivery Risk Prediction System.

The systematic methodology ensured high-quality data preparation and provided a strong foundation for building accurate predictive models.

CHAPTER 5

MACHINE LEARNING MODEL DEVELOPMENT AND EVALUATION

5.1 Introduction

Machine Learning is one of the most important technologies in Business Analytics because it enables organizations to identify hidden patterns within historical data and make accurate predictions for future events. In logistics and supply chain management, predictive models help organizations identify shipment delays before they occur, allowing managers to take preventive actions.

The objective of this chapter is to explain the machine learning algorithms used, the model development process, evaluation techniques, and the comparative analysis performed to identify the best-performing model for predicting late delivery risk.

5.2 Machine Learning Approach

Since the target variable (**Late_delivery_risk**) contains only two possible outcomes (Late Delivery and On-Time Delivery), the problem is classified as a **Binary Classification Problem**.

Several supervised learning algorithms were trained and evaluated using the same preprocessed dataset.

The workflow included:

- Data Preparation
 - Feature Engineering
 - Train-Test Split
 - Model Training
 - Model Evaluation
 - Model Comparison
 - Best Model Selection
-

5.3 Train-Test Split

The dataset was divided into training and testing subsets to evaluate model performance on unseen data.

The training dataset was used to build the machine learning models, while the testing dataset was used to measure prediction accuracy and generalization capability.

This approach reduces overfitting and ensures that the model performs effectively on new shipment records.

5.4 Logistic Regression

Logistic Regression was selected as the baseline classification algorithm because of its simplicity and interpretability.

Working Principle

Logistic Regression estimates the probability that a shipment belongs to the "Late Delivery" class using the logistic (sigmoid) function.

Advantages

- Easy to understand
- Fast training
- Low computational cost
- Good baseline model
- Interpretable coefficients

Limitations

- Assumes linear relationships
- Limited capability for complex nonlinear patterns

Despite its simplicity, Logistic Regression provides a strong benchmark for comparing more advanced ensemble algorithms.

5.5 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Each decision tree is trained using a different random subset of the data, and the final prediction is determined through majority voting.

Advantages

- High prediction accuracy
- Handles nonlinear relationships
- Robust against overfitting
- Effective with large datasets
- Handles mixed feature types

Random Forest is widely used in logistics because shipment delays are influenced by numerous interacting operational variables.

5.6 Gradient Boosting

Gradient Boosting improves predictive performance by sequentially building decision trees that correct the errors made by previous trees.

Instead of constructing trees independently, each new tree focuses on minimizing the prediction errors generated by earlier models.

Advantages

- High predictive accuracy
- Strong generalization
- Captures complex patterns
- Suitable for structured datasets

Gradient Boosting is particularly effective for business analytics problems involving multiple interacting variables.

5.7 XGBoost

Extreme Gradient Boosting (XGBoost) is one of the most powerful machine learning algorithms for structured tabular datasets.

XGBoost introduces several optimization techniques including regularization, parallel computation, and efficient handling of missing values.

Advantages

- Excellent accuracy
- Fast training
- Reduced overfitting
- Efficient memory utilization
- Strong scalability

Because of these characteristics, XGBoost has become one of the most widely used algorithms in predictive analytics competitions and industrial applications.

5.8 Model Training Process

The following workflow was followed during model development:

1. Load processed dataset.
2. Separate input features and target variable.
3. Apply preprocessing pipeline.
4. Split data into training and testing sets.
5. Train multiple classification models.
6. Evaluate model performance.
7. Compare evaluation metrics.
8. Select the best-performing model.
9. Save the trained model using Joblib.
10. Deploy the model within the Streamlit dashboard.

This systematic process ensured consistency across all algorithms and enabled fair performance comparison.

5.9 Model Evaluation Metrics

To compare the performance of the machine learning models, multiple evaluation metrics were used.

Accuracy

Accuracy measures the proportion of correctly classified shipments.

Formula

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision

Precision measures the percentage of predicted delayed shipments that were actually delayed.

A high Precision value indicates fewer false alarms.

Recall

Recall measures the ability of the model to correctly identify delayed shipments.

High Recall is important because missing a delayed shipment may result in operational problems.

F1 Score

The F1 Score is the harmonic mean of Precision and Recall.

It provides a balanced evaluation when both metrics are important.

ROC-AUC Score

ROC-AUC evaluates the model's ability to distinguish between delayed and non-delayed shipments across different classification thresholds.

Higher ROC-AUC values indicate better classification performance.

5.10 Model Comparison

The performance of all trained models was compared using multiple evaluation metrics.

Model Comparison Table

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.732661201	0.842841693	0.629874722	0.720959815	0.815583977
Random Forest	0.69369045	0.79960222	0.588957365	0.678302353	0.767607796
Gradient Boosting	0.713466652	0.887622016	0.546625581	0.676587364	0.774900654

XGBoost	0.715682473	0.885028 682	0.55334411	0.68094 3648	0.7829 54529
---------	-------------	-----------------	------------	-----------------	-----------------

5.11 Best Model Performance

Based on the evaluation results, the selected model achieved the following performance:

Evaluation Metric	Score
Accuracy	73.27%
Precision	84.28%
Recall	62.99%
F1 Score	72.10%
ROC-AUC	0.8156

These results demonstrate that the model provides reliable predictions and is capable of effectively identifying shipments at risk of late delivery.

5.12 Model Evaluation Visualizations

The following evaluation visualizations were generated during the project:

Figure 5.1

Classification Report

	precision	recall	f1-score	support
0	0.66	0.86	0.74	16308
1	0.84	0.63	0.72	19796
accuracy			0.73	36104
macro avg	0.75	0.74	0.73	36104
weighted avg	0.76	0.73	0.73	36104

Figure 5.2

Confusion Matrix

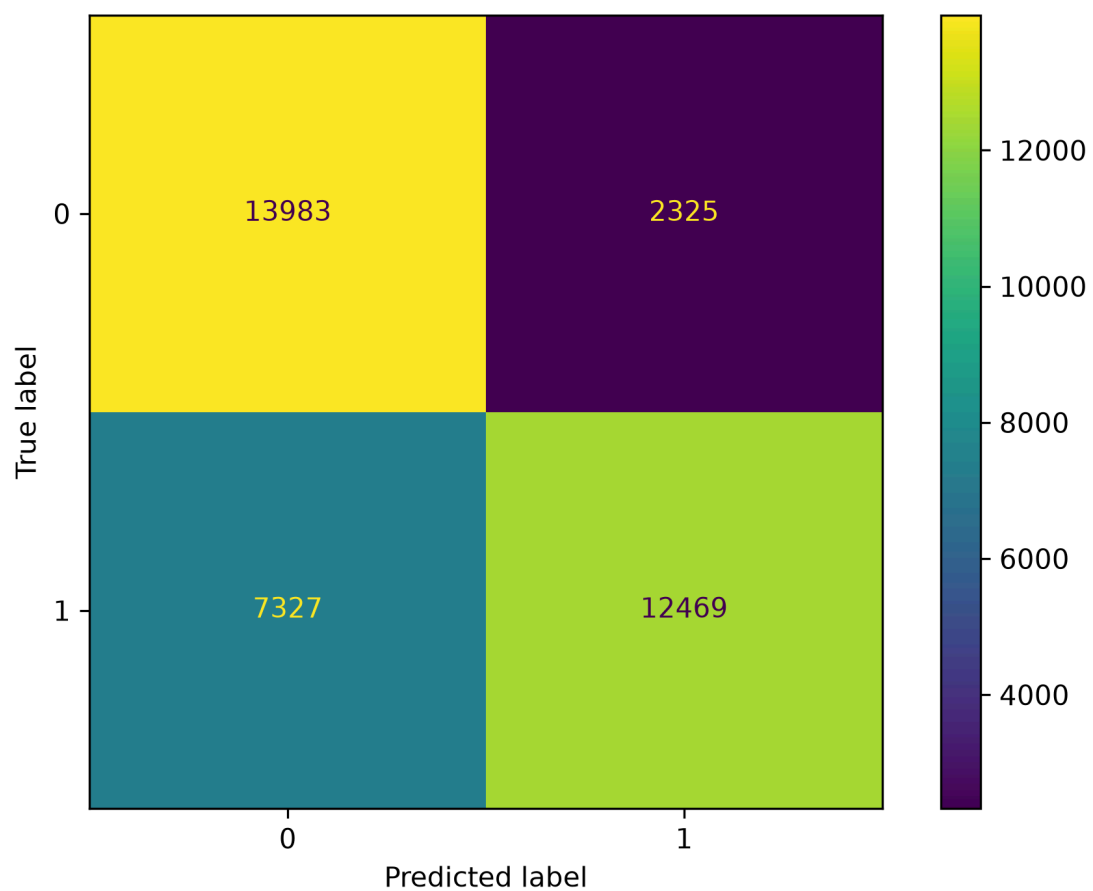


Figure 5.3

ROC Curve

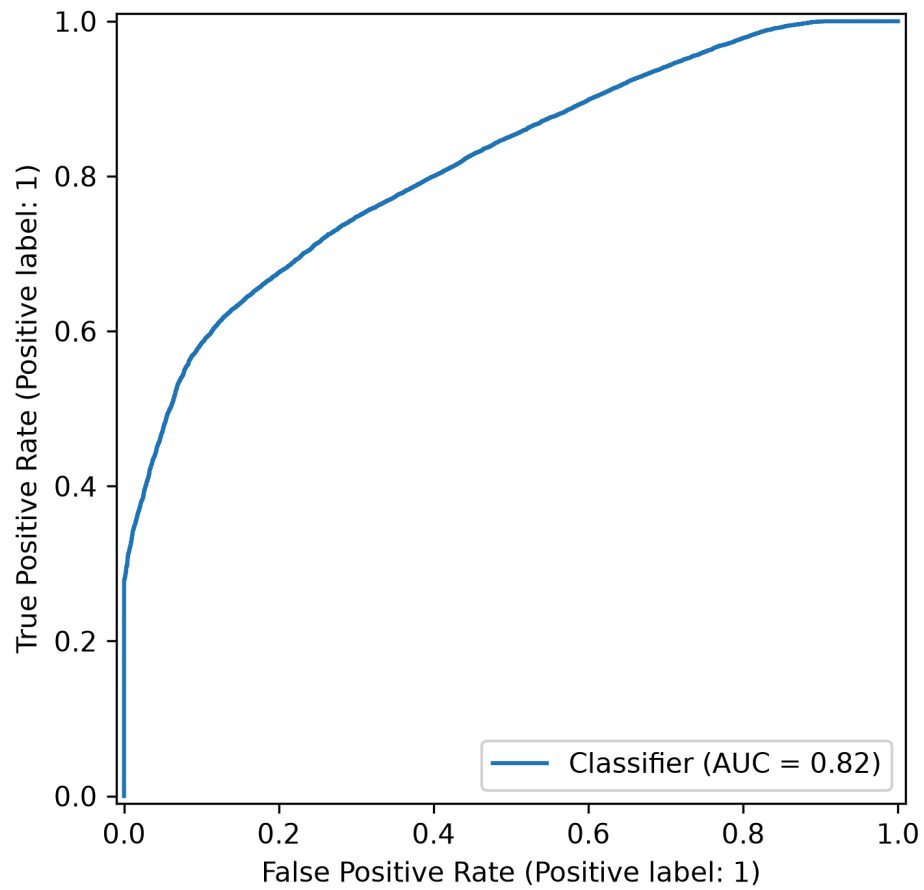
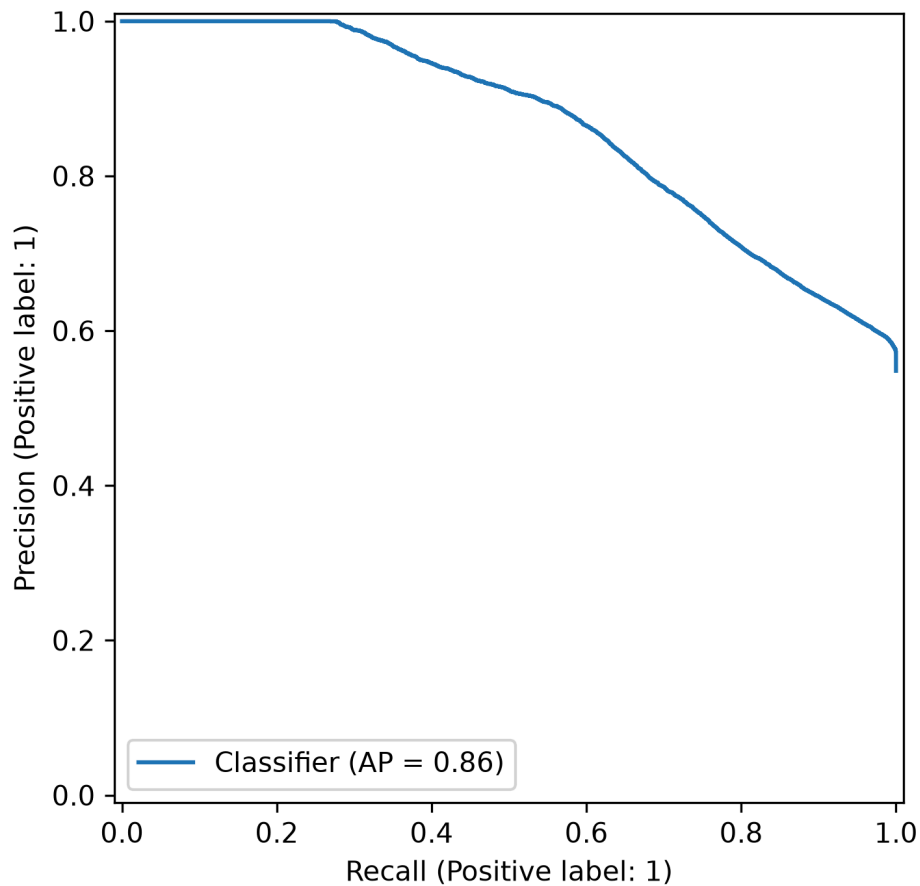


Figure 5.4

Precision–Recall Curve



5.13 Model Deployment

The final trained model was saved using the Joblib library and integrated into the Streamlit dashboard.

The deployment process included:

- Loading the trained model
- Loading the preprocessing pipeline
- Accepting user inputs
- Performing feature engineering
- Predicting late delivery risk
- Displaying prediction probability
- Categorizing shipment risk
- Providing business recommendations

The deployed dashboard enables logistics managers to perform predictive analysis without requiring programming knowledge.

5.14 Business Interpretation

The developed model provides valuable business insights that support operational decision-making.

The model enables organizations to:

- Predict shipment delays before dispatch.
- Identify high-risk orders.
- Improve shipment scheduling.
- Allocate resources more effectively.
- Improve customer communication.
- Reduce operational costs.
- Increase delivery performance.
- Enhance customer satisfaction.

These capabilities demonstrate the practical value of integrating machine learning into logistics operations.

5.15 Chapter Summary

This chapter presented the complete machine learning development process followed in this project. Multiple classification algorithms were trained and evaluated using standard performance metrics.

The comparative analysis enabled the selection of the best-performing model, which was successfully deployed within an interactive Streamlit dashboard to support predictive logistics analytics and business decision-making.

CHAPTER 6

STREAMLIT DASHBOARD DEVELOPMENT, CONCLUSION AND FUTURE SCOPE

6.1 Introduction

The final stage of this internship project involved deploying the trained Machine Learning model through an interactive web application. While predictive models generate valuable insights, they become significantly more useful when presented in a user-friendly interface that allows business users to interact with the predictions without requiring programming knowledge.

To achieve this objective, a multi-page interactive dashboard was developed using **Streamlit**, an open-source Python framework for building data applications. The dashboard enables logistics managers to explore shipment information, analyze risk trends, evaluate model performance, and identify high-risk shipments requiring immediate operational attention.

The deployment of the dashboard transforms the machine learning model into a practical business decision-support system.

6.2 Dashboard Development

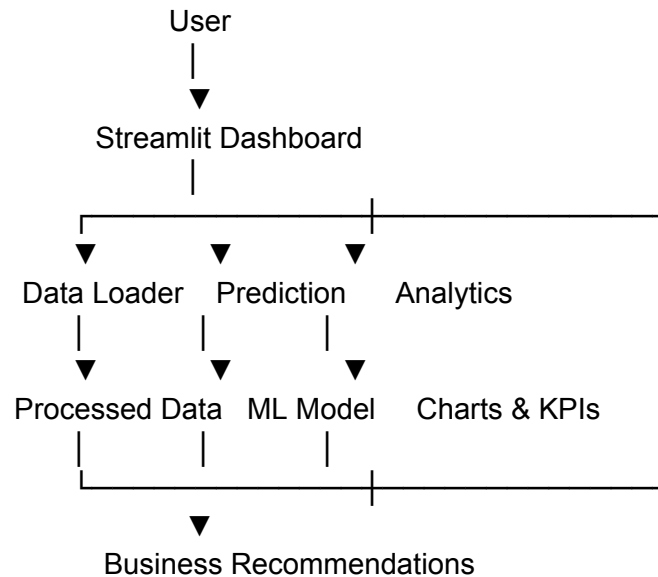
The dashboard was developed using the Streamlit framework due to its simplicity, flexibility, and ability to integrate seamlessly with Python-based machine learning models.

The application loads the trained model and preprocessing pipeline, accepts user inputs, performs feature engineering, generates risk predictions, and displays business insights using interactive visualizations.

The dashboard consists of multiple pages, each designed to support different aspects of logistics decision-making.

6.3 Dashboard Architecture

The overall architecture of the deployed solution is illustrated below.



The architecture integrates data loading, predictive modeling, visualization, and reporting into a unified application.

6.4 Dashboard Modules

Executive Dashboard

The Executive Dashboard provides a comprehensive overview of logistics operations through key performance indicators and interactive charts.

Features include:

- Total Shipments
 - High-Risk Shipments
 - Average Risk Probability
 - Shipping Mode Distribution
 - Customer Segment Analysis
 - Market Distribution
 - Regional Analysis
-

Risk Prediction

The Risk Prediction page allows users to predict the probability of late delivery for an individual shipment.

The page automatically performs feature engineering before passing the data to the trained machine learning model.

Displayed information includes:

- Prediction Result
 - Probability Score
 - Risk Category
 - Business Recommendation
-

Risk Analytics

The Risk Analytics page enables users to explore historical shipment trends and operational patterns.

Features include:

- Shipping Mode Analysis
- Market Analysis
- Customer Segment Distribution
- Regional Trends
- Interactive Filters

These visualizations help managers identify patterns associated with shipment delays.

Model Performance

This page presents detailed evaluation metrics for the trained machine learning model.

It includes:

- Model Comparison
- Confusion Matrix
- ROC Curve
- Precision–Recall Curve

- Classification Report

These metrics help users understand the predictive capability of the deployed model.

Operations Center

The Operations Center is designed to support operational decision-making by identifying shipments that require immediate attention.

Features include:

- Risk Threshold Slider
- High-Risk Shipment Table
- Risk Categories
- CSV Download
- Operational Recommendations

This module enables logistics managers to prioritize resources based on predicted shipment risk.

About Project

The About Project page provides documentation related to the project, including:

- Business Problem
- Objectives
- Dataset Information
- Technology Stack
- Machine Learning Workflow
- Business Value
- Future Scope

This section improves project transparency and serves as a reference for users.

6.5 Business Recommendations

Based on the results obtained during model development and dashboard analysis, the following recommendations are proposed:

1. Prioritize shipments identified as high risk by the prediction model.
2. Allocate additional logistics resources to regions with higher predicted delay probabilities.
3. Consider upgrading shipping methods for critical customer orders.
4. Monitor operational bottlenecks using the dashboard regularly.
5. Integrate predictive analytics into daily logistics planning.
6. Continuously retrain the machine learning model using updated shipment data.
7. Use dashboard insights to improve customer communication regarding potential delays.

These recommendations can help organizations improve delivery performance and operational efficiency.

6.6 Challenges Faced

During the development of this project, several technical and operational challenges were encountered:

- Handling a large logistics dataset containing over 180,000 records.
- Data preprocessing and feature engineering for mixed data types.
- Selecting the most suitable machine learning algorithm.
- Managing class imbalance during model evaluation.
- Debugging Streamlit dashboard integration.
- Resolving deployment issues related to file paths and repository structure.
- Optimizing the application for deployment on Streamlit Cloud.

Each challenge was addressed through systematic debugging, testing, and iterative improvements.

6.7 Project Outcomes

The project successfully achieved its objectives by delivering:

- An end-to-end machine learning solution for shipment delay prediction.
- A comparative evaluation of multiple classification algorithms.
- A trained predictive model with strong classification performance.

- A fully interactive Streamlit dashboard.
 - Business insights supporting proactive logistics management.
 - A publicly deployed application accessible through Streamlit Cloud.
 - A well-documented GitHub repository with complete project documentation.
-

6.8 Conclusion

This internship project successfully demonstrated the application of Business Analytics and Machine Learning in addressing the real-world challenge of predicting late delivery risk in global supply chain operations.

The project covered the complete analytics lifecycle, including data preprocessing, exploratory data analysis, feature engineering, predictive modeling, model evaluation, dashboard development, and cloud deployment.

By integrating machine learning with interactive business dashboards, the solution enables logistics managers to identify high-risk shipments before dispatch, optimize operational planning, and improve customer satisfaction.

The project also strengthened practical skills in Python programming, predictive analytics, data visualization, and business intelligence while providing valuable hands-on experience in solving industry-relevant problems.

Overall, the project highlights the potential of predictive analytics to enhance decision-making and operational efficiency in modern logistics organizations.

6.9 Future Scope

The project can be extended in several ways to further improve its capabilities:

- Real-time shipment monitoring.
- Integration with IoT-enabled logistics systems.
- REST API for enterprise applications.
- Cloud deployment using AWS, Azure, or Google Cloud Platform.
- Automated model retraining pipelines.
- Explainable AI using SHAP or LIME.
- Deep Learning-based prediction models.
- Integration with enterprise ERP systems.

- Mobile dashboard application.
- Email and SMS alerts for high-risk shipments.

These enhancements would make the solution more scalable and applicable to real-world logistics environments.

6.10 References

The following references were used during the development of this project:

Books

1. Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques*. Morgan Kaufmann.
2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning*. Springer.
3. Provost, F., & Fawcett, T. (2013). *Data Science for Business*. O'Reilly Media.